

CYBER FORENSIC DEPARTMENT
DIGITAL FORENSICS & INCIDENT RESPONSE (DFIR) UNITFORENSIC INVESTIGATION REPORT
INCIDENT ASSESSMENT: RANSOMWARE / EXTORTION

RESTRICTED - LAW ENFORCEMENT & SOC CONFIDENTIAL - CHAIN OF CUSTODY PRESERVED

Case Identifier:	CASE-2026-001	Report Version:	v2.5 Enterprise DFIR
Target Evidence:	ransomware_incident_syslog.log	Report Date:	18-Sep-2026 20:13:48 UTC+05:30
Incident Severity:	CRITICAL (100/100)	Investigation Status:	Active Containment / Triage Complete
Primary Analysts:	Pragati Patil, Tanisha Lohar, Gaurav Rajguru	Project Guide:	Dr. Shabanam K. Shikalgar
Department:	Department of Computer Science and Engineering (IoT & Cyber Security including Blockchain)	Institution:	ADCET, Ashta (Autonomous)

1. EXECUTIVE SUMMARY

The Cyber Forensic Department initiated an automated digital forensic triage of the evidence artifact '**ransomware_incident_syslog.log**' (2,646 bytes) to ascertain unauthorized activity, malicious payload execution, lateral movement, and data exfiltration. The automated triage engine classified the incident with an overall risk severity rating of **CRITICAL (Risk Score: 100/100)**.

Supervised Machine Learning classification (Random Forest) identified the primary threat vector as '**Ransomware / Extortion**' with **44.0% confidence**. The attack has advanced to Cyber Kill Chain stage: '**Critical Stage - Active Threat Impact / Ransomware Execution**' (100.0% progression). Unsupervised Anomaly Modeling (Isolation Forest) confirmed system deviation status as '**Anomalous**' (Anomaly Index: **0.00**).

Analysis discovered a cumulative total of **6 Indicators of Compromise (IOCs)**, **3 heuristic YARA signature matches**, and **5 correlated MITRE ATT&CK; techniques**. Immediate endpoint containment, credential revocation, and perimeter firewall blocking are required.

2. SCOPE AND METHODOLOGY

The scope of this forensic triage encompasses the examination of digital evidence submitted to the Cyber Triage platform. The analysis followed a forensically sound methodology compliant with ISO/IEC 27037 standards for digital evidence handling:

Evidence Artifact	SHA-256 Cryptographic Hash	Size	Format / Status
ransomware_incident_syslog.log	dd0be9cf2a87b54bed5cfee78b9f9dede56aa0b2419c69116f69fb1052a2f57d	2,646 B	LOG (Preserved)

- Cryptographic Verification:** Bit-level hashing (SHA-256, MD5, SHA-1) upon intake to ensure zero evidence tampering.
- Pattern & IOC Extraction:** Automated extraction of IPv4/IPv6, URLs, domains, hashes, CVEs, and Base64 payloads.
- YARA Heuristic Scanning:** Heuristic detection of ransomware strings, shadow copy deletion, Mimikatz, and webshells.
- Machine Learning Modeling:** Dual AI architecture combining Isolation Forest anomaly scoring with Random Forest classification.
- Threat Intelligence Enrichment:** Live reputation correlation against known threat actor infrastructures and C2 gateways.
- ATT&CK; & Kill Chain Mapping:** Normalization of forensic artifacts into MITRE Enterprise Tactics and Kill Chain stages.

3. AI ANOMALY & THREAT CLASSIFICATION RESULTS

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Analytical Metric	Quantitative Value	Forensic Interpretation
Overall Threat Risk Score	100 / 100	Calibrated severity level: CRITICAL
Supervised Threat Classifier (RF)	Ransomware / Extortion	Multi-class probability distribution: Benign / Normal Activity: 7.0% Ransomware / Extortion: 44.0% APT & C2 Communication: 10.0% Credential Theft & Access: 17.0% Web Application Compromise: 22.0%
Isolation Forest Anomaly Index	0.00 / 1.00	Unsupervised deviation status: Anomalous
Cyber Kill Chain Progression	100.0%	Lifecycle stage reached: Critical Stage - Active Threat Impact / Ransomware Execution
Shannon Binary Entropy	5.54 / 8.00	Measures code randomness, packing, or encryption density

4. MITRE ATT&CK; ENTERPRISE MAPPING

Tactic Name	Technique ID	Technique Name	Observed Artifact / Trigger
Impact	T1490	Inhibit System Recovery	vssadmin
Impact	T1486	Data Encrypted for Impact	lockbit
Execution	T1059.001	Command and Scripting Interpreter: PowerShell	powershell
Initial Access	T1190	Exploit Public-Facing Application	rce
Defense Evasion	T1070.001	Indicator Removal: Clear Windows Event Logs	wevtutil

5. INDICATORS OF COMPROMISE (IOCs)

5.1 Network Indicators (Defanged)

Type	Defanged Value	Threat Context / Reputation	Confidence
IP Address	185[.]220[.]101[.]5	Malicious C2 / TOR Exit Node LockBit 3.0 / APT29	95%
IP Address	194[.]26[.]29[.]112	Active Ransomware C2 Gateway LockBit Affiliate	90%
URL / Domain	hxxp://185[.]220[.]101[.]5/payload/stage2[.]e	Outbound HTTP Request	90%

5.2 Host & Endpoint Indicators

Type	Indicator / Artifact Path	Severity & Forensic Context
SHA-256	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855	Cryptographic Hash / Malware Sample
MD5	7b2c9a6234b071e62a8291a13426e95c	Cryptographic Hash / Malware Sample
LOLBAS Command	powershell.exe -nop -w hidden -enc	Severity: High (Living off the Land)
LOLBAS Command	vssadmin.exe delete shadows	Severity: Critical (Living off the Land)
LOLBAS Command	wmic shadowcopy delete	Severity: High (Living off the Land)

6. CHRONOLOGICAL INCIDENT TIMELINE

Timestamp (UTC)	Event Description	Category	Severity
2026-08-17 02:14:22	Aug 17 02:14:22 DC01-SRV Microsoft-Windows-Security-Aud	System Event	Info

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2026-08-17 02:14:45	Aug 17 02:14:45 DC01-SRV Microsoft-Windows-PowerShell[4]	Suspicious Execution / LOLBAS	High
2026-08-17 02:15:10	Aug 17 02:15:10 DC01-SRV Microsoft-Windows-Sysmon[1]: P	Ransomware / Critical Threat	Critical
2026-08-17 02:15:33	Aug 17 02:15:33 DC01-SRV Microsoft-Windows-Sysmon[1]: P	Ransomware / Critical Threat	Critical
2026-08-17 02:16:01	Aug 17 02:16:01 DC01-SRV Microsoft-Windows-Sysmon[1]: P	Ransomware / Critical Threat	Critical
2026-08-17 02:16:08	Aug 17 02:16:08 DC01-SRV Microsoft-Windows-Sysmon[1]: P	System Event	Info
2026-08-17 02:16:15	Aug 17 02:16:15 DC01-SRV Microsoft-Windows-Sysmon[1]: P	System Event	Info
2026-08-17 02:16:22	Aug 17 02:16:22 DC01-SRV Microsoft-Windows-Sysmon[1]: P	System Event	Info

7. YARA & HEURISTIC DETECTION SUMMARY

- [RULE-RANSOM-001] Ransomware Shadow Copy & Backup Invalidation (Critical Severity)
Details: Attempts to inhibit system recovery by deleting volume shadow copies or disabling startup repair.
- [RULE-RANSOM-002] Ransomware Note & Known Extortion Signatures (Critical Severity)
Details: Indicators of ransom notes, encryption extensions, or extortion group signatures (e.g. LockBit, Conti).
- [RULE-EVASION-001] Defense Evasion & Security Log Tampering (High Severity)
Details: Attempts to clear event logs, disable antivirus, or bypass execution policies.

8. CONTAINMENT, ERADICATION & RECOMMENDATIONS

8.1 Immediate Containment Actions (0–24 Hours):

- 8.1.1 [Network Containment] - Block 2 malicious external IP(s) on perimeter firewall and EDR.
- 8.1.2 [Endpoint Isolation] - Immediately disconnect host from corporate LAN and Wi-Fi to prevent lateral encryption spreading.
- 8.1.3 [Backup Verification] - Check offline immutable backups and verify integrity of VSS shadow copies on unaffected systems.
- 8.1.4 [Persistence Removal] - Audit and remove unauthorized Windows Registry Run keys, Scheduled Tasks, and anomalous startup services.

8.2 Short-Term Eradication (1–7 Days):

- Terminate unauthorized persistence mechanisms (Registry Run keys, scheduled tasks, and rogue services).
- Invalidate all compromised administrative credentials and force enterprise-wide Kerberos ticket resets.
- Re-image infected hosts using verified, immutable baseline golden images.

8.3 Long-Term Hardening & Post-Incident Security:

- Enforce hardware-backed Multi-Factor Authentication (MFA) across all administrative access points.
- Deploy Endpoint Detection & Response (EDR) agents with tamper protection and automated containment.
- Implement air-gapped, immutable offline backups to prevent ransomware impact.

9. CHAIN OF CUSTODY SUMMARY & CRYPTOGRAPHIC VERIFICATION

Timestamp	Action Performed	Performed By	Verification Hash / Block
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2026-08-18 00:02:14	Evidence Acquisition & Cryptographic Hashing	Gaurav Rajguru / Pragati Patil / Tanisha Lohar	SHA256: dd0be9cf2a87b54bed5cfee78b9f . . .
18-Sep-2026 20:13:48 UTC+05:30	AI Forensic Triage & Multi-Parser Extraction	Cyber Triage AI Core	Hash Chain: VERIFIED INTACT

10. LIMITATIONS AND ASSUMPTIONS

- 1. **Evidence Completeness:** Findings are strictly bounded by the digital artifacts provided for triage. Unacquired physical RAM or unallocated disk sectors may harbor auxiliary attacker footprints.
- 2. **Obfuscation & Encryption:** Adversary anti-forensic techniques (e.g. encrypted payloads, memory unhooking) may limit static string detection.
- 3. **Legal Admissibility:** Cryptographic hashes and custody logs have been maintained in compliance with Section 65B of the Indian Evidence Act.

11. AUTHORISATION & FORMAL SIGN-OFF

The undersigned forensic examiners and reviewing authorities hereby certify that this investigation was conducted under rigorous forensic protocols, and the findings presented herein represent an accurate, tamper-proof analysis of the evidence.

Role	Name & Academic / Professional Designation	Signature & Official Stamp	Date
Lead Forensic Examiners	Pragati Kiran Patil (23101024) Tanisha Prakash Lohar (23101009) Gaurav Subhash Rajguru (23101004) B.Tech CSE (IoT, Cyber Security & Blockchain)		18-09-2026
Reviewing Officer & Guide	Dr. Shabanam K. Shikalgar Guide, Dept. of CSE (IoT, CS & BT), ADCET Ashta		18-09-2026
Department Coordinator	Mr. Samish N. Kamble Project Coordinator, CSE [IOT-CSBT], ADCET Ashta		18-09-2026
Head of Department	Dr. Tahseen A. Mulla Head of Department, CSE [IOT-CSBT], ADCET Ashta		18-09-2026